

Regulatory and structural mechanisms of PvrA-mediated regulation of the PQS quorum-sensing system and PHA biosynthesis in *Pseudomonas aeruginosa*

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ABSTRACT

Pseudomonas aeruginosa is capable of causing acute and chronic infections in various host tissues, which depends on its abilities to effectively utilize host-derived nutrients and produce protein virulence factors and toxic compounds. However, the regulatory mechanisms that direct metabolic intermediates towards production of toxic compounds are poorly understood. We previously identified a regulatory protein PvrA that controls genes involved in fatty acid catabolism by binding to palmitoyl-coenzyme A (CoA). In this study, transcriptomic analyses revealed that PvrA activates the *Pseudomonas* quinolone signal (PQS) synthesis genes, while suppressing genes for production of polyhydroxyalkanoates (PHAs). When palmitic acid was the sole carbon source, mutation of *pvrA* reduced production of pyocyanin and rhamnolipids due to defective PQS synthesis, but increased PHA production. We further solved the co-crystal structure of PvrA with palmitoyl-CoA and identified palmitoyl-CoA-binding residues. By using *pvrA* mutants, we verified the roles of the key palmitoyl-CoA-binding residues in gene regulation in response to palmitic acid. Since the PQS signal molecules, rhamnolipids and PHA synthesis pathways are interconnected by common metabolic intermediates, our results revealed a regulatory mechanism that directs carbon flux from

carbon/energy storage to virulence factor production, which might be crucial for the pathogenesis.

INTRODUCTION

Pseudomonas aeruginosa is an opportunistic pathogen that predominates as a major cause of pulmonary infections in patients with cystic fibrosis (CF) and chronic obstructive pulmonary disease (COPD) (1–4). The successful colonization of *P. aeruginosa* depends on its capacity to orchestrate global gene expression in response to the host *in vivo* environment to evade immune clearance and efficiently utilize available local nutrients (5,6).

Pseudomonas aeruginosa produces a variety of virulence factors that inactivate host immune cells and cause tissue damage. The type III secretion system (T3SS) directly injects effector proteins into host cells, interfering with cell function or causing cell death (7,8). It has been demonstrated that *P. aeruginosa* preferentially injects T3SS effectors into neutrophils, a type of phagocyte that plays critical roles in host defence against bacterial pathogens (8). Besides the cell contact-dependent T3SS, *P. aeruginosa* produces multiple extracellular virulence factors, including exotoxin A, elastases LasA and LasB, alkaline protease AprA, phospholipase C and lipases (9–11). In addition, the secondary metabolites pyocyanin and rhamnolipids have been shown to play important roles in the pathogenesis of *P. aeruginosa* (12,13). Pyocyanin is a redox-active molecule that provokes oxidative damages in host cells, leading to cell death (12,14,15). The amphipathic rhamnolipids are involved in degradation of lung surfactant, disruption of epithelial tight

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junctions and induction of rapid necrosis of neutrophils (13,16–18).

Production of pyocyanin and rhamnolipids is largely regulated by the quorum-sensing systems, namely the Las, Rhl and *Pseudomonas* quinolone signal (PQS) systems (19). The autoinducer molecules of the Las and Rhl systems are *N*-3-oxododecanoyl-L-homoserine lactone (C12HSL) and *N*-butyryl-L-homoserine lactone (C4HSL), which bind to and activate the cognate transcriptional regulators LasR and RhlR, respectively (20–22). In the PQS system, the transcriptional regulator PqsR (MvfR) binds to the PQSs 3,4-dihydroxy-2-alkylquinoline and 4-hydroxy-2-heptylquinoline (HHQ), and activates the transcription of the *pqsABCDE*, *phnAB* operons and *pqsH*, that encode enzymes involved in the syntheses of PQSs and HHQ, thus forming a positive feedback loop (23–25). Meanwhile, the PQS molecule has been shown to act as a mediator in outer membrane vesicle generation, iron acquisition, cytotoxicity and modulation of host immune responses (23).

Previous studies demonstrated that long chain fatty acids (LCFAs) are one of the major carbon sources that are utilized by *P. aeruginosa* during lung infection (26–28). The exogenous LCFAs are converted to fatty acyl-CoA by fatty acyl-CoA synthetase (FadD), followed by degradation through the β -oxidation cycle (27). Mutations in the fatty acid degradation genes attenuated the bacterial virulence during mouse lung infection (28). In addition, LCFAs have been shown to function as signalling molecules that induce expression of virulence genes in pathogenic bacteria, including *Mycobacterium tuberculosis*, *Helicobacter pylori*, *Vibrio cholera* and *Salmonella enterica* (29–33).

Previously, we identified a transcriptional regulator PvrA that binds to palmitoyl-CoA and regulates genes involved in the utilization of LCFAs. In this study, we examined the role of PvrA in regulating virulence gene expression in response to LCFAs. Our results demonstrate that PvrA controls production of pyocyanin and rhamnolipids through the PQS system in response to palmitic acid, while suppressing the synthesis of the carbon and energy storage compound polyhydroxyalkanoates (PHAs). This study reveals a novel regulatory mechanism that promotes the bacterial infection in response to host nutrients.

MATERIALS AND METHODS

Bacterial strains, plasmids and culture media

The bacterial strains and plasmids used in this study are listed in Supplementary Table S1. Bacterial cells were grown in Luria–Bertani (LB) broth (10 g/l tryptone, 5 g/l yeast extract, 5 g/l NaCl, pH 7.4). Medium with sole carbon source was prepared in 1× M9 medium (49.4 mM Na₂HPO₄, 24.0 mM KH₂PO₄, 9 mM NaCl, 19 mM NH₄Cl, 0.5 mM MgSO₄) with 0.2% (w/v) Brij-58, supplemented with 0.1% (w/v) glucose or 0.4% (w/v) palmitic acid, resulting in Glu-M9 or FA-M9 (26–28,34). The Fa-M9 medium is FA-M9 without Brij-58, and the FR-M9 medium is FA-M9 supplemented with rhamnolipids.

Transcriptome sequencing

Wild-type PA14 and the $\Delta pvrA$ mutant were cultured in FA-M9 at 37°C and harvested at the log phase (OD₆₀₀ = 1.0). Total RNA was isolated with an RNeasy pure bacteria kit (Qiagen Biotech, Beijing, China). Transcriptome sequencing was performed by Azena Life Sciences (Suzhou, China). A Ribo-Zero rRNA Removal Kit (Illumina, San Diego, CA, USA) was used to selectively remove rRNA. The ribosomal-depleted mRNA was then fragmented and reverse-transcribed. First-strand cDNA was synthesized using ProtoScript II Reverse Transcriptase with random primers and actinomycin D. The second-strand cDNA was synthesized using the Second Strand Synthesis Enzyme Mix (include dACGTP/dUTP). The double-stranded DNA was purified by the AxyPrep Mag PCR Clean-up (Axygen) and then treated with End Prep Enzyme Mix for both end repair and addition of a dA-tailing in one reaction, followed by a T–A ligation to add adaptors to both ends. After adenylation of the 3' ends of the DNA fragments, Illumina PE adaptor oligonucleotides were ligated to prepare for hybridization. DNA fragments 300 bp in length with adaptor molecules ligated on both ends were enriched using the Illumina PCR Primer Cocktail. The library was then sequenced on a NextSeq 500 platform (Illumina). The sequence reads were mapped onto the PA14 reference genome (NC_002516.2).

Reverse transcription and quantitative polymerase chain reaction

cDNA was synthesized with a reverse transcriptase and random primers (TAKARA, Dalian, China) from the total RNA. Quantitative reverse transcription–polymerase chain reaction (qRT–PCR) was performed by mixing the indicated specific primers, cDNA and a SYBR Premix Ex Taq II™ (TaKaRa) in a total volume of 20 μ l with three biological replicates. *rpsL*, the gene encoding the 30S ribosomal protein, was used as an internal control.

Extraction and quantification of PHA

Extraction of PHA was performed as previously described with minor modifications (35). Bacterial cultures (30 ml) in Glu-M9 or FA-M9 medium were centrifuged at 10 000 *g* for 5 min. Then the pellets were homogenized with commercial sodium hypochlorite solution (6–14% active chlorine basis, MACKLIN) at room temperature for 1 h. After centrifugation for 40 min at 12 000 *g*, the pellets were washed with double-distilled water and precipitated with 1:1 alcohol–acetone. The pellets were resuspended in chloroform and incubated for 48 h at 25°C. Then, the content of PHA was determined as chrotonic acid in H₂SO₄ by detecting absorbance at 235 nm (36,37).

Gas chromatography/mass selective detection (GC/MSD; Agilent) was used to analyse the composition and yield of the PHA (35,38). The GC analysis was performed as follows: column, HP-5MS (Agilent); initial, 80°C for 8 min; heating rate, 8°C/min up to 150°C and 2°C/min up to 176°C; holding, 176°C for 1 min; 1 μ l of the sample was injected at 250°C; carrier (He) flow rate, 34.1

ml/min; oven temperature, 250°C; detector temperature, 300°C.

Quantification of PQS

The relative PQS level was measured by using a PAO1 $\Delta pqsA$ mutant containing a *pqsA-gfp* transcriptional fusion on a plasmid ($\Delta pqsA/PpqsA-gfp$) as the reporter strain (39). Cultures of the indicated strains were centrifuged at 14 000 g for 5 min. The supernatant was collected and filtered through a 0.2 μ m filter. The reporter strain $\Delta pqsA/PpqsA-gfp$ was grown overnight and adjusted to an OD₆₀₀ of 0.2. A 100 μ l aliquot of the filtered supernatant was mixed with 100 μ l of the $\Delta pqsA/PpqsA-gfp$ cells and inoculated into the well of a 96-well plate. The green fluorescent protein (GFP) fluorescence was measured with a fluorometer (Varioskan Flash, Thermo Scientific) and normalized to OD₆₀₀.

Quantification of pyocyanin

The pyocyanin level was determined as previously described with minor modifications (40). Briefly, overnight bacterial culture in LB was diluted 100-fold into the indicated fresh medium and grown for 48 h. A 0.6 ml aliquot of chloroform was added to 1 ml of supernatant of the bacterial culture. After centrifugation at 13 000 g for 5 min, the organic layer was carefully collected and mixed with 0.2 ml of HCl (0.2 N). The mixture was centrifuged at 13 000 g for 1 min. The absorbance at 520 nm of the upper liquid was then measured with a fluorometer (Varioskan Flash; Thermo Scientific).

Quantification of rhamnolipids

The rhamnolipid levels were determined as previously described (41). Bacteria were grown overnight in LB at 37°C, and then diluted 1:200 in 50 ml of fresh FA-M9 medium and grown at 37°C for 24 h. The supernatant was collected after centrifugation at 13 000 g for 5 min and the pH was adjusted to 2.0 with HCl. Then 30 ml of the supernatant was mixed with an equal volume of chloroform-methanol (2:1), followed by shaking at 200 rpm for 30 min. The mixture was centrifuged at 9000 g for 15 min. A 100 μ l aliquot of the lower organic phase was mixed with 2 ml of ddH₂O, 1 ml of phenol and 5 ml of sulphuric acid (98%), followed by incubation at 80°C for 1 h. After centrifugation at 10 000 g for 1 min, the absorbance at 490 nm of the supernatant was measured. The values were substituted into the standard curve of rhamnose ($y = 0.042x - 0.035$, $R^2 = 0.9586$) to obtain the concentration of rhamnose. The concentration of rhamnolipids was obtained by multiplying the concentration of rhamnose by the glycolipid coefficient 3.4.

Cytotoxicity assay

The bacterial cytotoxicity was determined by the lactate dehydrogenase (LDH) release assay. A total of 2×10^5 A549 cells were seeded into each well of a 24-well tissue culture plate, and cultured in Dulbecco's modified Eagle medium (DMEM) with 10% (v/v) thermally inactivated

foetal bovine serum, streptomycin (100 mg/ml) and penicillin G (100 U/ml) at 37°C with 5% (v/v) CO₂ at 37°C for 18–24 h. Before infection, the cell culture medium was replaced by DMEM with 2.5% bovine serum albumin (BSA). PA14, the $\Delta pvrA$ mutant and the $\Delta pvrA/pvrA$ strain were cultured at 37°C in the indicated medium to an OD₆₀₀ of 1.0 and then the bacterial cells were washed twice in phosphate-buffered saline (PBS) before infecting the cells at a multiplicity of infection (MOI) of 50. The LDH level in the medium was determined with the LDH cytotoxicity assay kit (Beyotime, Shanghai, China) at 2 h post-infection.

To test the cytotoxicity of the bacterial culture supernatant, bacteria were cultured at 37°C overnight in the indicated medium. The supernatants were collected after centrifugation at 12 000 g, followed by filtration through a 0.22 μ m filter. Then 100 μ l of the supernatant was added to the cells, followed by determination of LDH release after 10 h.

Animal experiment

All the animal assays in this study were carried out in accordance with National guidelines on the use of animals in research. The protocols were approved by the animal care and use committee of Nankai University, College of Life Sciences (Permission number: NK-04-2012).

The murine acute pneumonia model was performed as previously described with minor modifications (42). In brief, overnight bacterial cultures were grown to an OD₆₀₀ of 1.0 at 37°C in LB. The bacteria were collected by centrifugation at 12 000 g for 1 min, followed by washing with PBS twice. The bacteria were then resuspended in PBS to a concentration of 2×10^8 colony-forming units (CFU)/ml. Female BALB/c mice at 6–8 weeks old were randomly divided into two groups (six mice per group). After being anaesthetized through intraperitoneal injection of 80–100 μ l of 7.5% chloral hydrate, each mouse was intranasally inoculated with 20 μ l of the bacterial suspension. At 6 hours post-infection (hpi), the mice were sacrificed by CO₂. Bacteria in the bronchi alveolar lavage fluid (BALF) were collected by centrifugation at 12 000 g for 2 min. The total RNA was isolated using Trizol Reagent (Thermo Fisher Scientific, USA).

β -Galactosidase activity assay

Bacteria were grown in LB overnight and diluted 1:100 in the indicated medium. After 4 h, 500 μ l of the bacterial culture was collected and resuspended in 1.5 ml of Z buffer (0.01 M KCl, 0.001 M MgSO₄, 0.06 M Na₂HPO₄, 0.04 M NaH₂PO₄ and 0.05 M β -mercaptoethanol). A 1 ml aliquot of the bacterial suspension was used for OD₆₀₀ measurement. The remaining 500 μ l of suspension was mixed with 10 μ l of chloroform and 10 μ l of 0.1% sodium dodecyl sulphate (SDS) by vortex. Then 100 μ l orthonitrophenyl-galactopyranoside (ONPG, 4 mg/ml) (BBI life sciences) was added to the mixture, followed by incubation at 37°C. When the colour of the mixture turned light yellow, 500 μ l of Na₂CO₃ (1 M) was added to stop the reaction, followed by measurement of OD₄₂₀. The β -galactosidase activities were calculated as: Miller units = $(1000 \times OD_{420}) / (T / 500 / OD_{600})$ (T, reaction time).

Electrophoretic mobility shift assay (EMSA)

EMSA experiments were performed as described previously with minor modifications (26). The indicated DNA fragments (200 ng) were incubated with purified wild-type or mutated PvrA protein on ice for 20 min and then at 37°C for 5 min in a 20 µl reaction buffer (4% glycerol, 10 mM pH 7.6 Tris-HCl, 5 mM CaCl₂, 100 mM NaCl, 1 mM EDTA and 10 mM β-mercaptoethanol) with or without palmitoyl-CoA (10⁻³ µg/µl). The electrophoresis was performed on a 5% native polyacrylamide gel in Tris-borate-EDTA (TBE) buffer at 100 V for 1 h. Bands were stained with ethidium bromide (0.5 µg/ml) and visualized with the molecular imager ChemiDoc™ XRS + (Bio-Rad).

5'-Rapid amplification of cDNA ends (RACE) assay

Total RNA was extracted from PA14 grown in FA-M9 medium, followed by reverse transcription with a reverse transcriptase and random primers (Takara, Dalian, China). Poly(dG) was added to the 3' end of the purified cDNA by using a terminal deoxynucleotidyl transferase (Takara, Dalian, China). The resultant cDNA was used as template for PCR amplification with primers *phnA*-RaceR1 or *phaG*-RaceR1 together with 5-AP (Supplementary Table S1). The PCR products were used as templates for another round of PCR with primers *phnA*-RaceR2 or *phaG*-RaceR2 together with 5-NP (Supplementary Table S1). The amplified products were purified by agarose gel electrophoresis, and then sequenced to determine the transcription start site (TSS).

Expression and purification of PvrA

The *pvrA* (PA2957) gene was PCR amplified from *P. aeruginosa* PA14 genomic DNA using the forward primer 5'-CGGATCCCAGAAAGAGCCTCGCAAAGTT-3' and the reverse primer 5'-CCCCGGGCACACAGGCATCAGG-3'. The *pvrA* gene was cloned into the pGEX-6P-1 vector, and the recombinant plasmid was transformed into the *Escherichia coli* strain BL21(DE3) and grown in LB medium containing 100 µg/ml ampicillin at 37°C. When the OD₆₀₀ reached 0.6–0.8, isopropyl-β-D-1-thiogalactopyranoside (IPTG) was added to LB medium at a final concentration of 0.3 mM to induce the expression of the recombinant protein. The bacteria were then grown for 16–18 h at 16°C. The bacterial cells were harvested by centrifugation at 4000 rpm for 20 min at 16°C and resuspended in a lysis buffer (20 mM Tris, 150 mM NaCl, pH 9.0). The bacterial cells were lysed by sonication on ice. The cell debris was removed by centrifugation at 18 000 rpm for 40 min at 4°C. The supernatant was gradually added to the glutathione *S*-transferase (GST) column that had been equilibrated with the lysis buffer. Once all the supernatant had been loaded into the column, the GST column was washed extensively with a washing buffer (20 mM Tris, pH 9.0). The PreScission protease was then added to the resin to remove the GST tag. The protein was eluted with 20 mM Tris, pH 9.0 and the eluate was concentrated to 1 ml. The protein was further purified by anion exchange chromatography on an equilibrated HiTrap Q column (GE Healthcare). The eluate was concentrated to 1 ml by Amicon Ultracel-10K

(Millipore). The concentrated protein sample was then loaded onto a Superdex 200 column (GE Healthcare) that had been equilibrated with 20 mM Tris, 150 mM NaCl, pH 9.0.

For the preparation of selenomethionyl (SeMet)-derivative PvrA protein, cells were grown overnight at 37°C in 5 ml of LB medium containing 100 µg/ml ampicillin. Cells from overnight cultures were collected by centrifugation and resuspended in 1 ml of M9 medium (7 g/l Na₂HPO₄, 3 g/l KH₂PO₄, 1 g/l NH₄Cl, 0.5 g/l NaCl). Cell resuspensions were used to inoculate 1 l of M9 minimal medium supplemented with 1 mM MgSO₄, 0.1 mM CaCl₂, 0.4% glucose and 100 µg/ml ampicillin, and grown at 37°C until the OD₆₀₀ reached 0.4–0.6. The following L-amino acids were then added: Lys, Phe and Tyr (100 mg/l), Leu, Ile, Thr and SeMet (50 mg/l). Protein expression was induced by the addition of 0.3 mM IPTG at an OD₆₀₀ of 0.6–0.8. The cultures were then incubated overnight at 16°C. Dithiothreitol (2 mM) was added to the lysis buffer used for SeMet-PvrA protein purification and crystallization. SeMet-PvrA was purified following the same procedure as above using the pGEX-6P-1 vector. All purified proteins were concentrated to 10 mg/ml, flash-frozen with liquid nitrogen and stored at –80°C for later use.

Crystallization and data collection

Crystallization of PvrA and SeMet-derivative PvrA (SeMet-PvrA) was performed using the sitting-drop vapour diffusion method at 20°C with the Index and Wizard Screen™ kits (Hampton Research). By mixing equal volumes of protein solution (5 mg/ml) and crystallization solution, we observed regular and appropriately sized crystals after 1 week. PvrA crystals were grown at a protein concentration of 5 mg/ml from a condition containing 10% w/v polyethylene glycol (PEG) 1000 and 10% w/v PEG 8000. SeMet-PvrA crystals with a protein concentration of 10 mg/ml were obtained from a condition containing 25% w/v PEG 3350, pH 7.5. Crystallization of the PvrA-palmitoyl-CoA complex was achieved by co-crystallization of PvrA with 3 mM palmitoyl-CoA, and crystals were grown from a condition containing 30% w/v PEG 4000, 0.2 M magnesium chloride hexahydrate, 0.1 M Tris-HCl, pH 8.0. Prior to data collection, crystals were immersed in the crystallization buffer supplemented with 25% glycerol and immediately flash-cooled in liquid nitrogen.

High-resolution X-ray data (Supplementary Table S2) were collected at 100 K at Shanghai Synchrotron Radiation Facility (SSRF) on beamlines 18U1 (PvrA, PvrA-palmitoyl-CoA) and 19U1 (SeMet-PvrA). All diffraction images were integrated, scaled and merged with the HKL2000 software package (43).

Structure determination

The structure of PvrA was determined by the single-wavelength anomalous dispersion (SAD) method using SeMet derivative data. A total of 16 heavy atom sites were identified and phases were calculated using AutoSol (44) in the PHENIX package (45). Most of the structure in two

monomers was built automatically using AutoBuild (46) in the PHENIX package, and the remainder of the structure was manually built using Coot. Refinement of the structure in PHENIX was alternated with cycles of manual adjustment in Coot (47). The native and palmitoyl-CoA structures were determined by molecular replacement in PHENIX using the PvrA-SeMet structure as a search model and refined using the same protocol. The quality of the final structures was assessed using MolProbity (48). Structure figures were drawn using PyMOL.

Isothermal titration calorimetry (ITC)

The ITC assay was performed using a MicroCal iTC200 (Malvern) as previously described (26). The ligand palmitoyl-CoA (0.2 mM) was injected into the PvrA protein or the point mutants (0.02 mM) at 25°C in a reaction buffer (20 mM Tris-HCl, 100 mM NaCl, pH 7.5). The apparent heat change was determined after each injection and analysed with the software ORIGIN.

RESULTS

PvrA reversely regulates PQS and PHA biosynthesis genes upon utilizing palmitic acid

To understand the global regulatory role of PvrA in response to LCFA, we performed transcriptome analyses after growing wild-type PA14 and the $\Delta pvrA$ mutant in an M9 minimal medium with palmitic acid as the sole carbon source (FA-M9) (27). Compared with wild-type PA14, 111 genes and 50 genes were up- and down-regulated in the $\Delta pvrA$ mutant, respectively (Figure 1A; Supplementary Figure S1; Supplementary Table S3). Among the alternatively expressed genes, the *pqsABCDE* and *phnAB* operons that are involved in the biosynthesis of the PQS signal molecule were down-regulated, whereas the PHA biosynthesis gene *phaG* was up-regulated in the $\Delta pvrA$ mutant (Table 1). qRT-PCR results verified the expression patterns of *pqsA*, *pqsE*, *phnA* and *phaG* in the $\Delta pvrA$ mutant in FA-M9, which was restored by complementation with a *pvrA* gene (Figure 1B). In contrast, when glucose was the sole carbon source (Glu-M9), the expression levels of *pqsA*, *pqsE* and *phnA* were similar in wild-type PA14 and the $\Delta pvrA$ mutant (Figure 1B). The expression of *phaG* was up-regulated in the $\Delta pvrA$ mutant in Glu-M9, but to a lesser extent than in FA-M9 (Figure 1B).

We next examined the production levels of PHA and the PQS signal molecules. By using a spectrophotometric assay, we found that mutation of *phaG* resulted in an ~25% reduction in the PHA production in FA-M9 (Figure 2A). In agreement with the gene expression patterns, the $\Delta pvrA$ mutant produced 3.3-fold more PHA than the wild-type PA14 in FA-M9, whereas no significant difference was observed in Glu-M9 (Figure 2A). To further verify the results, we analysed the isolated PHA by GC-mass spectrometry (MS). When grown in FA-M9, 3-hydroxydecanoate, 3-hydroxydodecanoate and 3-hydroxytridecanoate were the three most abundant PHA subunits produced by the bacteria (Supplementary Figure S2A) and the $\Delta pvrA$ mutant produced 5.6-fold more total PHA than PA14, which was

restored to the wild-type level by complementation with a *pvrA* gene (Supplementary Figure S2B, C).

The relative amounts of the PQS signal molecules were measured by using a report system (39). When grown in FA-M9, the $\Delta pvrA$ mutant secreted fewer PQS signal molecules than wild-type PA14 (Figure 2B). However, the PQS signal molecules produced by the two strains were similar when the bacteria were grown in Glu-M9 (Figure 2B). Taken together, these results demonstrate that PvrA activates the PQS system and represses PHA synthesis in response to palmitic acid.

PvrA controls the production of pyocyanin and rhamnolipids through the PQS system

The PQS system controls production of pyocyanin and rhamnolipids that are toxic to mammalian cells and play important roles in the pathogenesis of *P. aeruginosa* (12,13,17,23,25). In FA-M9, the pyocyanin and rhamnolipid biosynthesis genes (*phzA*, *rhlA* and *rhlB*) were down-regulated in the $\Delta pvrA$ mutant (Figure 3A). In agreement with the gene expression patterns, the $\Delta pvrA$ mutant produced less pyocyanin and rhamnolipids than the wild-type PA14 (Figure 3B, C; Supplementary Figure S3A). The supernatant of the $\Delta pvrA$ mutant was less toxic to A549 cells than that of the wild-type PA14 (Figure 3D), whereas the bacterial cell-mediated cytotoxicities were similar (Supplementary Figure S3B). Growth in Glu-M9 resulted in similar expression levels of the pyocyanin and rhamnolipid biosynthesis genes in wild-type PA14 and the $\Delta pvrA$ mutant. Meanwhile, the amounts of pyocyanin and rhamnolipids as well as the cytotoxicity of the supernatants of the two strains were similar (Figure 3A–D).

Rhamnolipids possess excellent emulsifying power with a variety of hydrophobic substrates (49,50). Thus, besides cytotoxicity, rhamnolipids might facilitate absorption of fatty acids by *P. aeruginosa*. In FA-M9 without the emulsifier Brij-58 (designated as Fa-M9), mutation of the *rhlA* or *rhlB* gene reduced the bacterial growth rate, which was restored by supplementation of exogenous rhamnolipids (Supplementary Figure S4). As for the $\Delta pvrA$ mutant, supplementation of rhamnolipids increased the bacterial growth rate in Fa-M9 (Supplementary Figure S4). These results reveal a role for PvrA in regulating the production of rhamnolipids in response to palmitic acid which facilitates utilization of the fatty acid.

We next examined whether the down-regulation of the pyocyanin and rhamnolipid synthesis genes in the $\Delta pvrA$ mutant is due to the defective PQS system. Addition of exogenous PQS in FA-M9 increased the expression of *pqsA*, *phnA*, *phzA*, *rhlA* and *rhlB* in the $\Delta pvrA$ mutant (Figure 4A). In addition, the production of pyocyanin by the $\Delta pvrA$ mutant was increased by the exogenous PQS (Figure 4B; Supplementary Figure S5).

Since palmitic acid is abundant in the lung (51), we examined whether PvrA exerts a similar regulatory role during lung infection. Mutation of the *pvrA* gene increased the expression of *phaG*, while it reduced the expression of *pqsA*, *phnA*, *phzA*, *rhlA* and *rhlB* (Figure 5), indicating a role for PvrA in regulating these genes *in vivo*.

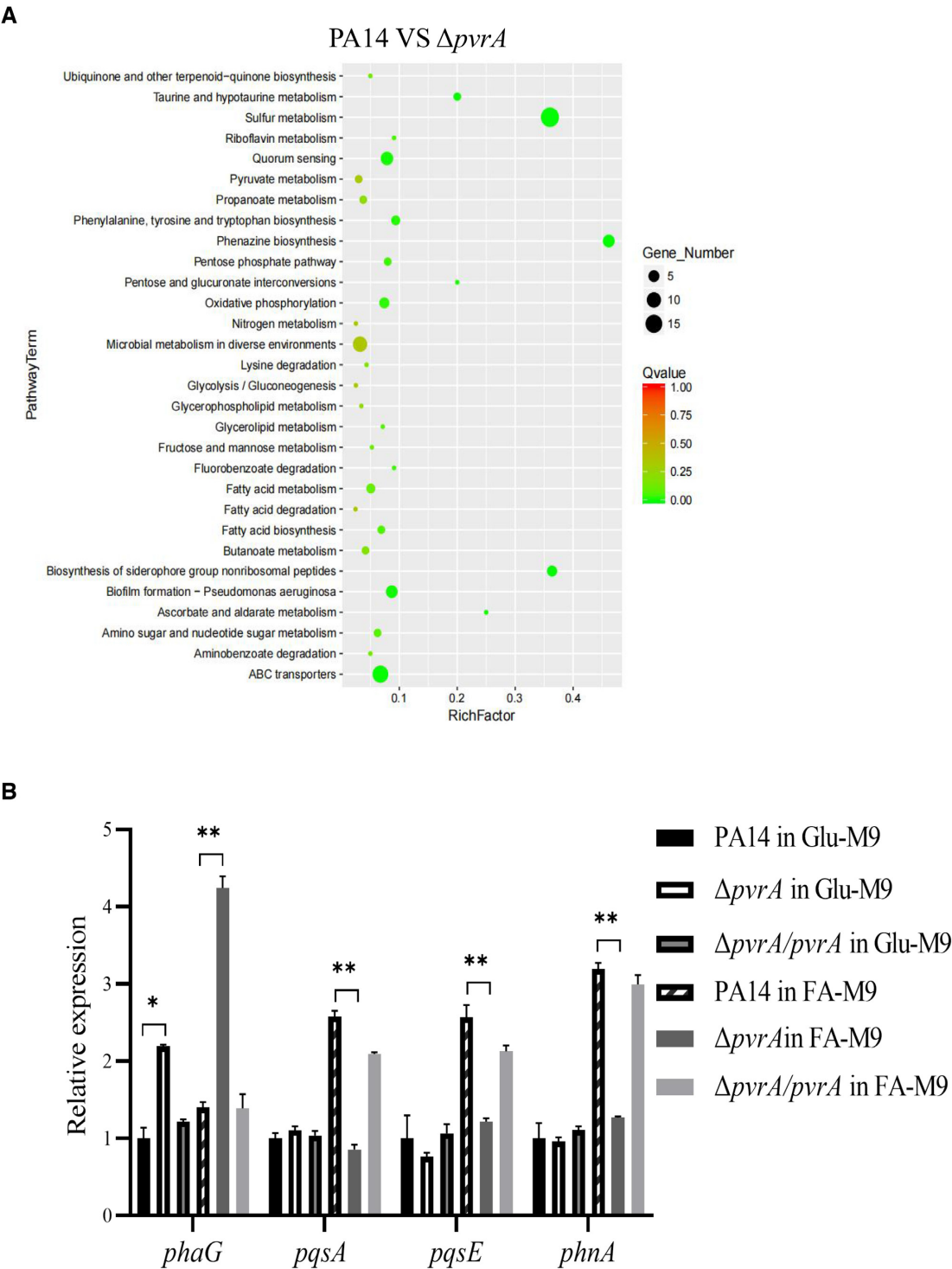
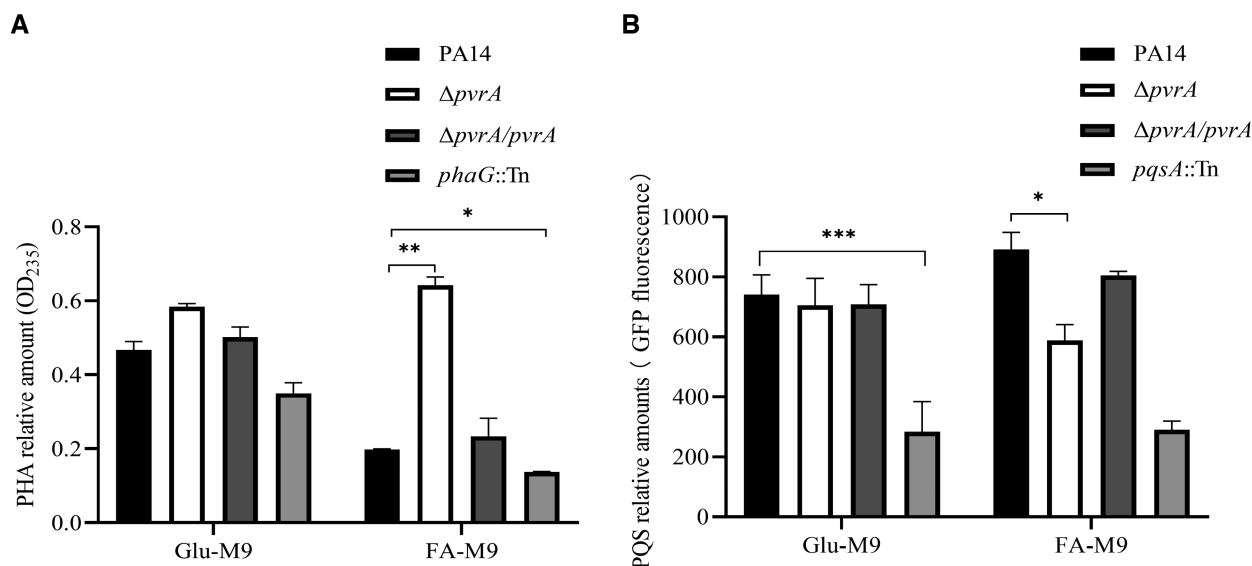


Figure 1. Transcriptome analysis of wild-type PA14 and the $\Delta pvrA$ mutant grown in FA-M9. (A) KEGG enrichment analysis of differentially expressed genes. The size and colour of each dot represent the number of differentially expressed genes and the Q-value ranges, respectively. (B) Wild-type PA14 and the $\Delta pvrA$ and $\Delta pvrA/pvrA$ strains were grown in Glu-M9 or FA-M9 to an OD₆₀₀ of 1.0. The relative mRNA levels of *phaG*, *pqsA*, *pqsE* and *phnA* were determined by qRT-PCR. **P* < 0.05; ***P* < 0.01 by Student's *t*-test. Data represent the mean $\pm$ standard deviation (SD) from three samples.

Table 1. Differently expressed PQS and PHA synthesis genes in the $\Delta pvrA$ mutant when grown in FA-M9

Gene ID	Gene name	Associated pathway	Description	Fold change ($\Delta pvrA$ /PA14)
PA0730	<i>phaG</i>	PHA biosynthesis	(R)-3-Hydroxydecanoyl-ACP:CoA transacylase	30.4
PA0996	<i>pqsA</i>	PQS biosynthesis	Anthranilate-CoA ligase	0.195
PA0997	<i>pqsB</i>	PQS biosynthesis	Hypothetical protein	0.200
PA0998	<i>pqsC</i>	PQS biosynthesis	Hypothetical protein	0.212
PA0999	<i>pqsD</i>	PQS biosynthesis	3-Oxoacyl-ACP synthase	0.204
PA1000	<i>pqsE</i>	Quorum-sensing regulation	Thioesterase PqsE	0.205
PA1001	<i>phnA</i>	Anthranilate synthesis	Anthranilate synthase component I	0.177
PA1002	<i>phnB</i>	Anthranilate synthesis	Anthranilate synthase component II	0.157

**Figure 2.** PHA and PQS production in Glu-M9 or FA-M9. (A) Wild-type PA14 and the strains $\Delta pvrA$, $\Delta pvrA/pvrA$ and *phaG*::Tn were grown in Glu-M9 or FA-M9. Relative amounts of the extracted PHA were determined by measuring the absorbance at 235 nm. (B) PA14 and the strains $\Delta pvrA$, $\Delta pvrA/pvrA$ and *pqsA*::Tn were grown in Glu-M9 or FA-M9. The PQS levels in the supernatants were determined by a reporter strain. ** $P < 0.01$; *** $P < 0.001$ by Student's *t*-test. Data represent the mean $\pm$ SD from three samples.

PvrA directly regulates PHA and quinolone biosynthesis genes

Our previous results revealed that PvrA binds to palmitoyl-CoA and recognizes a predicted consensus sequence 5'-CGGTCA-3' or the complementary sequence 5'-TGACCG-3' (26). The motif was identified in the promoter regions of *pqsA*, *phnA* and *phaG* (Figure 6A). *pqsA* contains two TSSs at 339 and 71 bp upstream of the start codon (52). The RhlR- and PqsR-binding sites are shown in Figure 6A (52). 5'-RACE assays revealed that the TSSs of *phnA* and *phaG* are at 187 and 35 bp upstream of the start codons, respectively (Figure 6A).

To examine whether PvrA directly controls the expression of these genes, we firstly constructed transcriptional fusions between *lacZ* and the promoters of these genes. Mutation of *pvrA* resulted in up-regulation of the P_{phaG} -*lacZ* and down-regulation of P_{pqsA} -*lacZ* and P_{phnA} -*lacZ* fusions, which was restored by complementation with a *pvrA* gene (Figure 6B). Removal of the predicted PvrA-binding sites from the *lacZ* transcriptional fusions increased the expression of P_{phaG} -*lacZ* and reduced the expression of P_{pqsA} -*lacZ* and P_{phnA} -*lacZ* in wild-type PA14, resulting in similar LacZ levels between wild-type PA14 and the $\Delta pvrA$ mutant (Figure 6A, B).

EMSAs and competitive EMSAs confirmed the direct binding of PvrA to the promoter regions of *pqsA*, *phnA* and *phaG* (Figure 6C; Supplementary Figure S6A) but not of *pvrA* (Supplementary Figure S6B). In addition, palmitoyl-CoA enhanced the binding of PvrA to the promoter regions of *phaG*, *pqsA* and *phnA* (Figure 6D; Supplementary Figure S6C).

Crystal structure of PvrA and insights into palmitoyl-CoA binding

To understand the structural basis for palmitoyl-CoA binding by PvrA and how it enhances promoter binding, we determined the high-resolution crystal structure of PvrA to 2.3 Å and PvrA (SeMet) to 2.1 Å. Data collection and refinement statistics are shown in Supplementary Table S2. The PvrA and PvrA (SeMet) structures are essentially identical and can be superimposed with a root mean square deviation (r.m.s.d.) of 0.6 Å for all residues. PvrA is a dimer both in the crystal and in solution, as evidenced by analytical ultracentrifugation which shows a single monodisperse peak at ~45 kDa corresponding to twice the theoretical molecular mass of 25 kDa (Supplementary Figure S7). Each monomer adopts a typical TetR fold with nine α -helices separated into two domains: an N-terminal DNA-

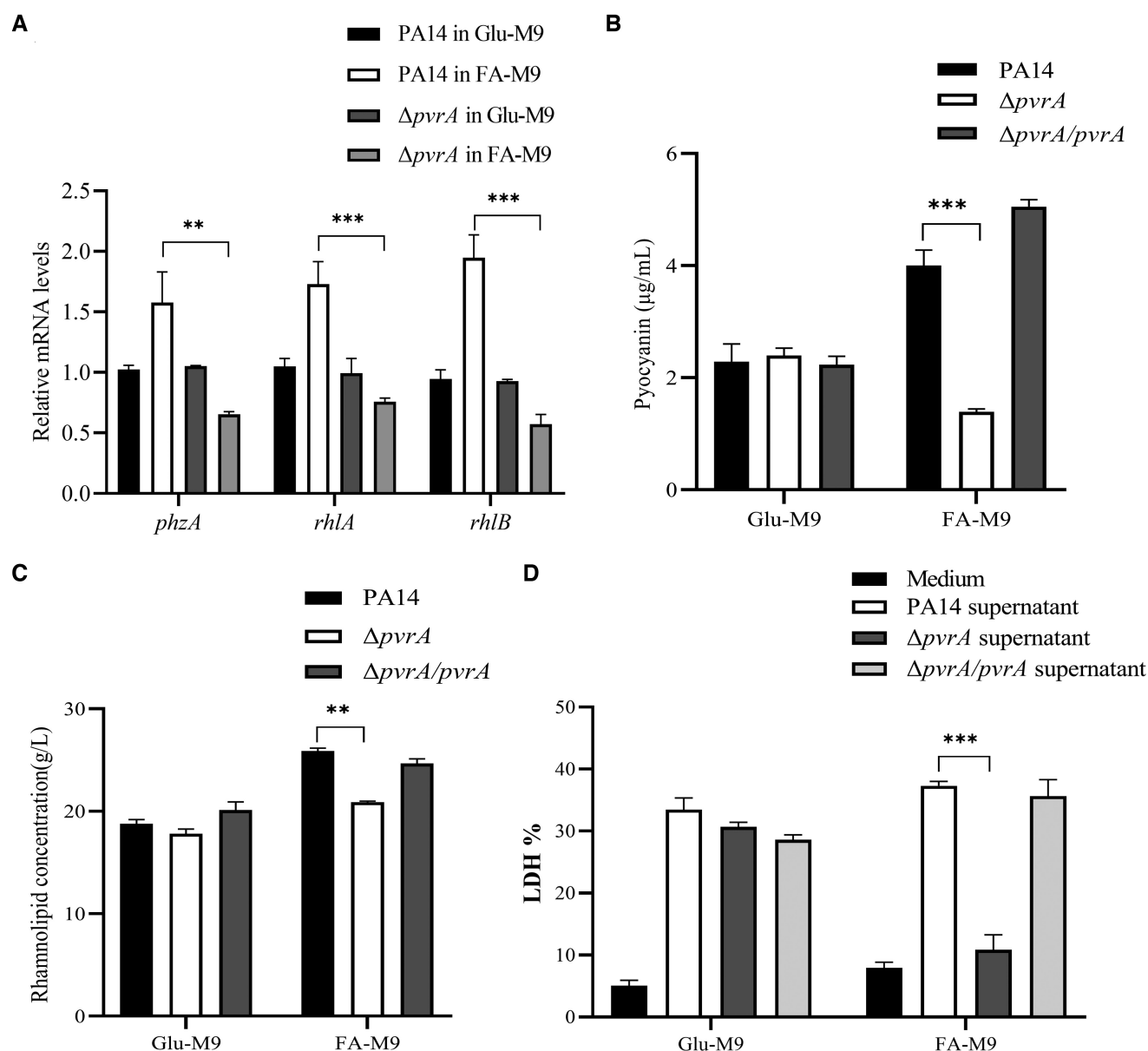


Figure 3. Roles of PvrA in regulating production of pyocyanin and rhamnolipids. (A) Wild-type PA14 and the $\Delta pvrA$ mutant were grown to an OD₆₀₀ of 1.0 in Glu-M9 or FA-M9. The relative mRNA levels of *phzA*, *rhlA* and *rhlB* were determined by qRT-PCR. (B) Wild-type PA14 and the $\Delta pvrA$ and $\Delta pvrA/pvrA$ strains were grown for 48 h in Glu-M9 or FA-M9, and the extracellular pyocyanin concentrations were measured. (C) Wild-type PA14 and the $\Delta pvrA$ and $\Delta pvrA/pvrA$ strains were grown overnight, and rhamnolipid concentrations were measured. (D) The bacteria were grown overnight in Glu-M9 or FA-M9. The same amounts of the culture supernatants were added to the A549 cells. The cytotoxicity was determined by the LDH release assay. LB medium was used as a negative control. ** $P < 0.01$; *** $P < 0.001$ by Student's *t*-test. Data represent the mean $\pm$ SD from three samples.

binding domain consisting of helices $\alpha 1$ – $\alpha 3$, in which helices $\alpha 2$ and $\alpha 3$ form the helix–turn–helix (HTH) motif, and a C-terminal ligand-binding domain consisting of helices $\alpha 4$ – $\alpha 9$ (Fig 7A). Dimerization occurs via a four α -helix bundle formed by helices $\alpha 8$ and $\alpha 9$ of each monomer (Figure 7A).

To elucidate the structural basis for palmitoyl-CoA, a co-crystal structure of PvrA with palmitoyl-CoA was determined. Clear electron density enabled the building of the adenosine 3',5'-diphosphate and pantothenic acid groups of CoA, while no interpretable density was observed for the palmitoyl group. The resulting structure shows that

CoA binds in the C-terminal domain in a 'side entry' mode (53), with the palmitoyl group predicted to extend into the space between the monomers (Figure 7B). The CoA complex structure can be superimposed onto the native structure with an r.m.s.d. of 0.8 Å for all residues, indicating no major conformational changes upon palmitoyl-CoA binding. In this structure, CoA primarily interacts with residues on helices $\alpha 7$ and $\alpha 8$ of the ligand-binding domain, as well as with residues on helices $\alpha 5$, $\alpha 6$ and the $\alpha 8$ – $\alpha 9$ turn of the neighbouring monomer (Figure 7B, C). Key residues that interact with palmitoyl-CoA include: Y92, F134, L137, Y159, C160 and W163 which form a pocket that accom-

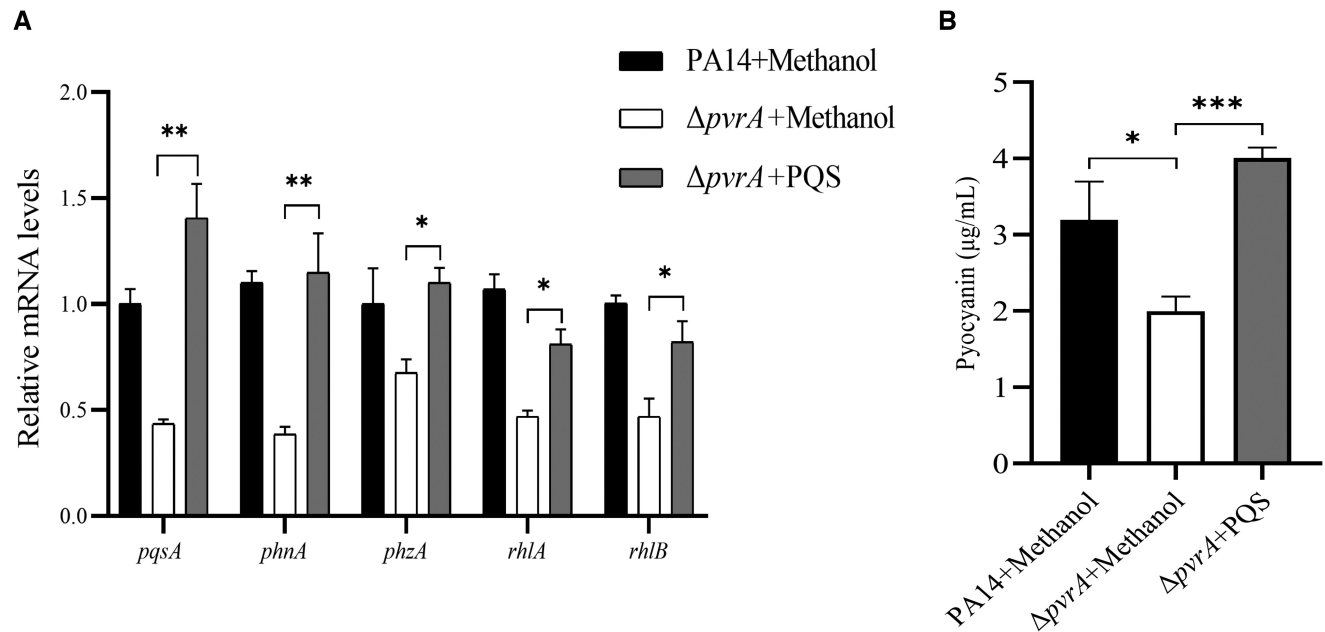


Figure 4. PvrA regulates pyocyanin and rhamnolipid production through PQS. (A) Wild-type PA14 and the $\Delta pvrA$ mutant were grown in FA-M9 with PQS or the solvent methanol to an OD₆₀₀ of 1.0. The relative mRNA levels of *pqsA*, *phzA*, *rhIA* and *rhIB* were determined by qRT-PCR. (B) Wild-type PA14 and the $\Delta pvrA$ mutant were grown in FA-M9 with PQS or the solvent methanol for 48 h. The extracellular pyocyanin concentrations were measured. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ by Student's *t*-test. Data represent the mean $\pm$ SD from three samples.

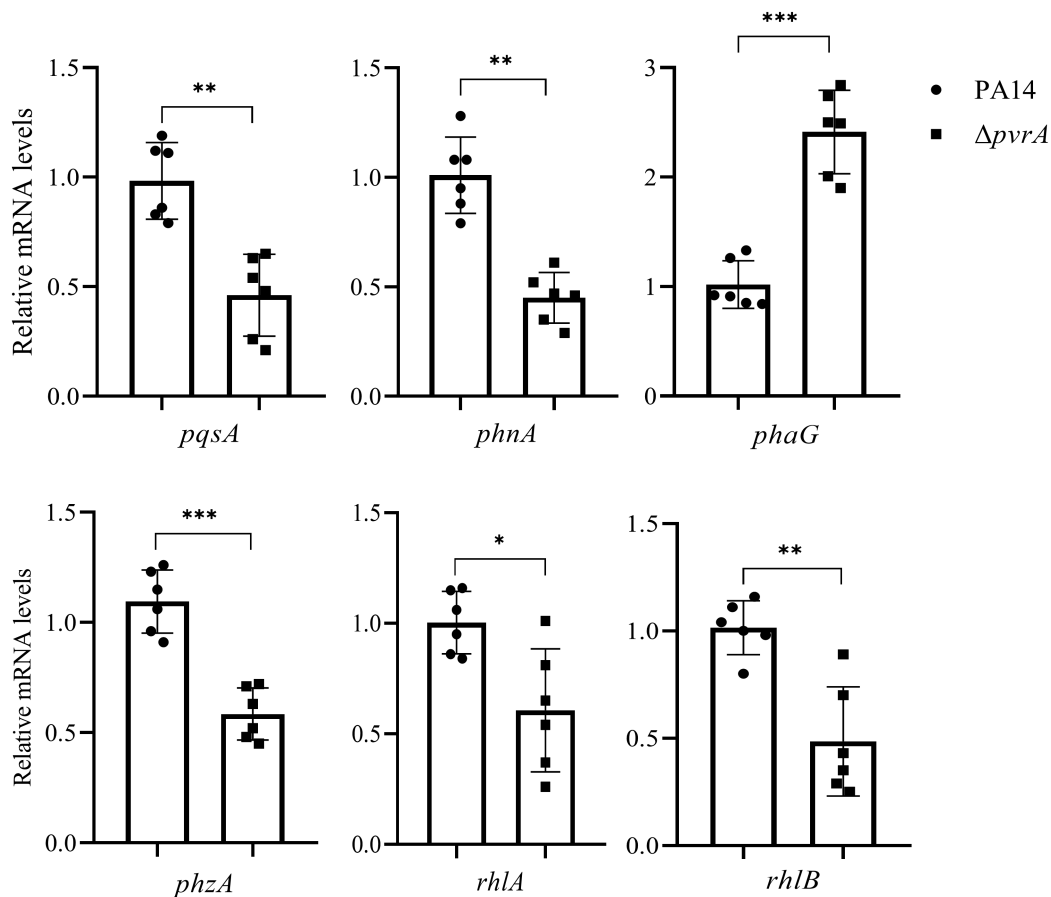


Figure 5. Regulatory role of PvrA during lung infection. Wild-type PA14 and the $\Delta pvrA$ mutant were grown in LB to an OD₆₀₀ of 1.0. Each mouse was inoculated with the bacteria intranasally. The bacteria were isolated from the BALF of each mouse at 6 hpi. The mRNA levels of the indicated genes were determined by qRT-PCR. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ by Student's *t*-test. Data represent the mean $\pm$ SD from six samples.

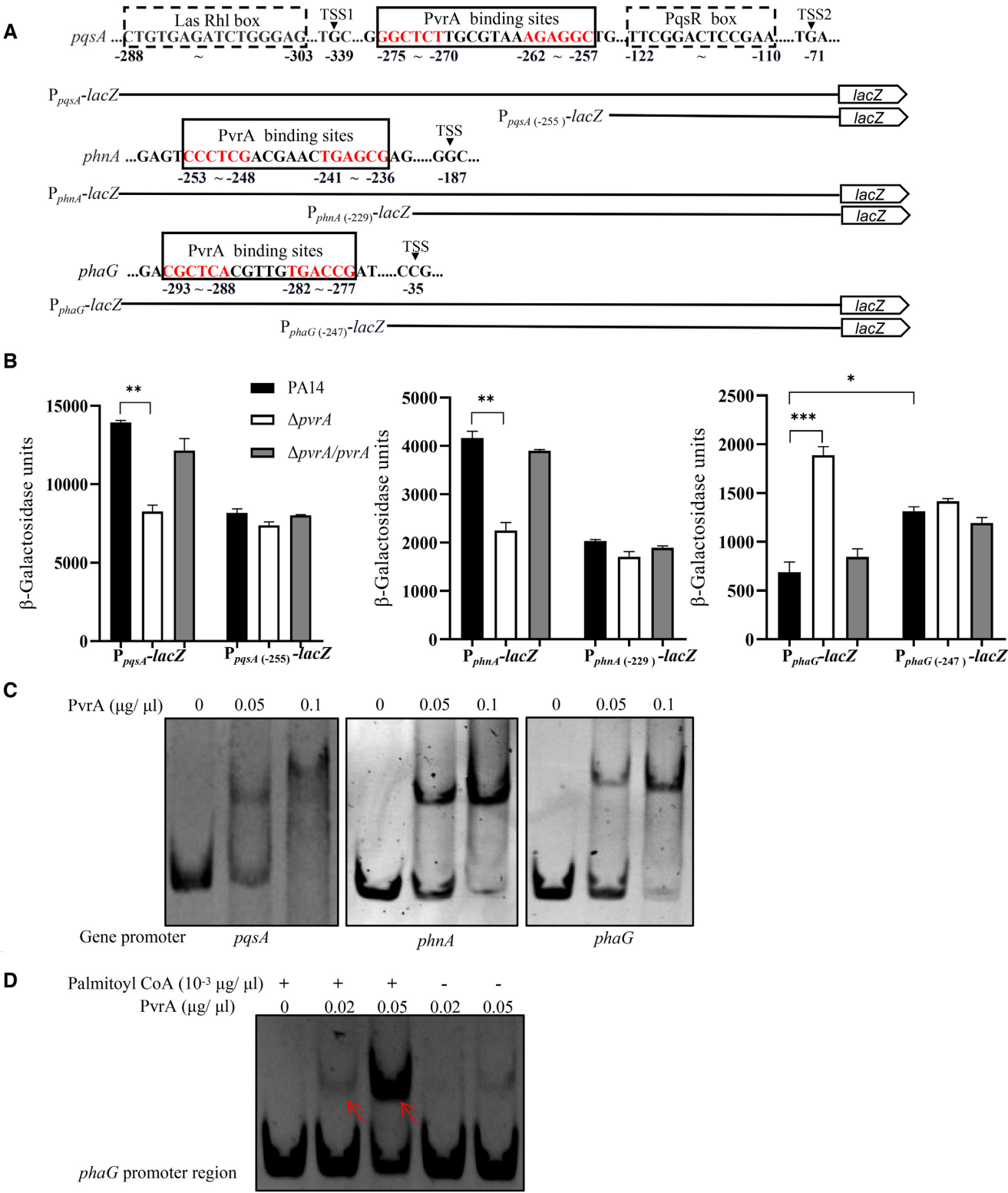


Figure 6. PvrA directly regulates PQS and PHA synthesis genes. (A) Potential PvrA-binding sites in the promoter regions of *pqsA*, *phnA* and *phaG* are shown in red and boxed with solid lines. The binding sites of RhlR and PqsR are boxed with dotted lines. The TSSs are indicated by arrowheads. The numbers indicate the distance upstream of the start codon. Schematic diagrams of the P_{pqsA} -*lacZ*, $P_{pqsA(-255)}$ -*lacZ*, P_{phnA} -*lacZ*, $P_{phnA(-229)}$ -*lacZ*, P_{phaG} -*lacZ* and $P_{phaG(-247)}$ -*lacZ* transcriptional fusions are shown below the individual promoter regions. (B) Bacteria carrying the indicated *lacZ* transcriptional fusions were grown in FA-M9 to an OD₆₀₀ of 1.0, followed by β-galactosidase assays. ***P* < 0.01; ****P* < 0.001 by Student's *t*-test. Data represent the mean ± SD from three samples. (C) Binding of PvrA to the promoter regions of the indicated genes was examined by EMSA. (D) Palmitoyl-CoA enhances the binding of PvrA to the *phaG* promoter region. The *phaG* promoter fragment was incubated with increasing concentrations of PvrA with or without the indicated concentrations of palmitoyl-CoA. The palmitoyl-CoA-enhanced PvrA–DNA interaction is indicated by red arrows.

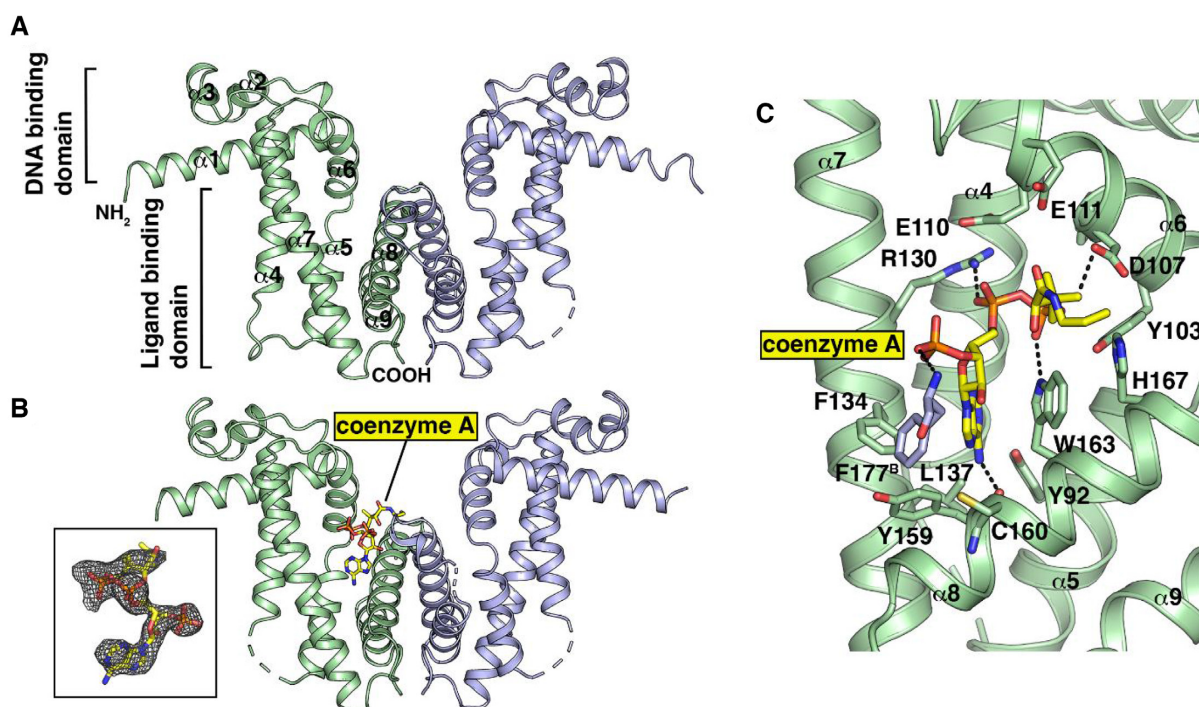


Figure 7. Structural analysis of PvrA. (A) Crystal structure of the native PvrA dimer. The structure is shown in cartoon representation with monomer A coloured pale green and monomer B coloured pale blue. Secondary structure elements and the amino (NH₂) and carboxyl (COOH) termini are labelled on monomer A. (B) Crystal structure of the PvrA–CoA complex. The structure is shown in cartoon representation with monomer A coloured pale green and monomer B coloured pale blue. The CoA ligand is shown in yellow stick representation. The inset shows CoA covered by 2mF_o–DF_c electron density calculated with the ligand omitted, contoured at 1.2 σ . (C) Enlarged view of the co-crystal structure of PvrA with palmitoyl-CoA showing interacting residues. The view shown is rotated 90° clockwise about the y-axis relative to (A). The CoA ligand is shown in yellow stick representation. Monomer A is shown in cartoon representation and coloured pale green. Interacting residues from monomer A are shown in green stick representation. Ser175 (Ser175^B) and Phe177 from monomer B (Phe177^B) are shown in blue stick representation. Hydrogen bonds are denoted by black dashed lines.

modates the adenine moiety; F177 of the neighbouring monomer which interacts with the phosphate group of the 3'-phosphoadenosine moiety; Y103, E110, R130 and W163 which interact with the diphosphate group; and D107, E111 and H167 which interact with the pantothenic acid group together with S175 of the neighbouring monomer (Figure 7C).

ITC was then used to validate the binding model by mutating residues predicted to interact with palmitoyl-CoA (Figure 8). A solution without protein was tested as a negative control (Supplementary Figure S8). Ten point mutations, namely Y92A, D107A, E110A, E111A, R130A, L137A, Y159A, W163A, H167A and F177A, reduced the binding affinity for palmitoyl-CoA by >100-fold (Figure 8). EMSA results demonstrated that palmitoyl-CoA did not enhance or slightly enhanced the binding between these mutants and the DNA probe (Figure 9A). Expression of each of the *pvrA* mutants in the $\Delta pvrA$ mutant resulted in similar expression levels of *phaG*, *pqsA*, *phnA* and *phzA* and pyocyanin production to those in the $\Delta pvrA$ mutant (Figure 10A, B; Supplementary Figure S9), which agrees with their low affinities for palmitoyl-CoA. The C160A mutation reduced the binding affinity of PvrA for palmitoyl-CoA or the DNA probe (Figures 8 and 9), and resulted in slightly higher *phaG* expression and lower *pqsA*, *phnA* and *phzA* expression and pyocyanin production (Figure 10A, B; Supplementary Figure S9). The Y103A and F106A mutations

reduced the binding affinity of PvrA for palmitoyl-CoA by >800-fold (Figure 8). However, the F106A mutant constitutively binds to the DNA probe with a higher affinity, and the Y103A mutant exhibited supershifted bands with or without palmitoyl-CoA (Figure 9). Expression of the F106A or Y103A mutant in the $\Delta pvrA$ mutant reduced the expression of *phaG* and increased the expression of *pqsA*, *phnA* and *phzA* and pyocyanin production (Figure 10A, B; Supplementary Figure S9). The R96A and F134A mutations reduced the binding affinities of PvrA for palmitoyl-CoA by ~30- and 60-fold, respectively (Figure 8). Palmitoyl-CoA enhanced the binding between R96A or F134A and the DNA probe (Figure 9). Expression of the F106A or Y103A mutant in the $\Delta pvrA$ mutant restored the expression levels of *phaG*, *pqsA*, *phnA* and *phzA* and pyocyanin production (Figure 10A, B; Supplementary Figure S9), indicating that these two residues might not play an important role in the function of PvrA.

DISCUSSION

Here in this study, we demonstrated that PvrA reversely regulates the PQS quorum-sensing system and biosynthesis of PHAs in response to palmitic acid in *P. aeruginosa*. In human lung, phosphatidylcholine (PC) makes up ~80% of the surfactant lipid (54), which has been shown to be a major nutrient for *P. aeruginosa* during lung infection

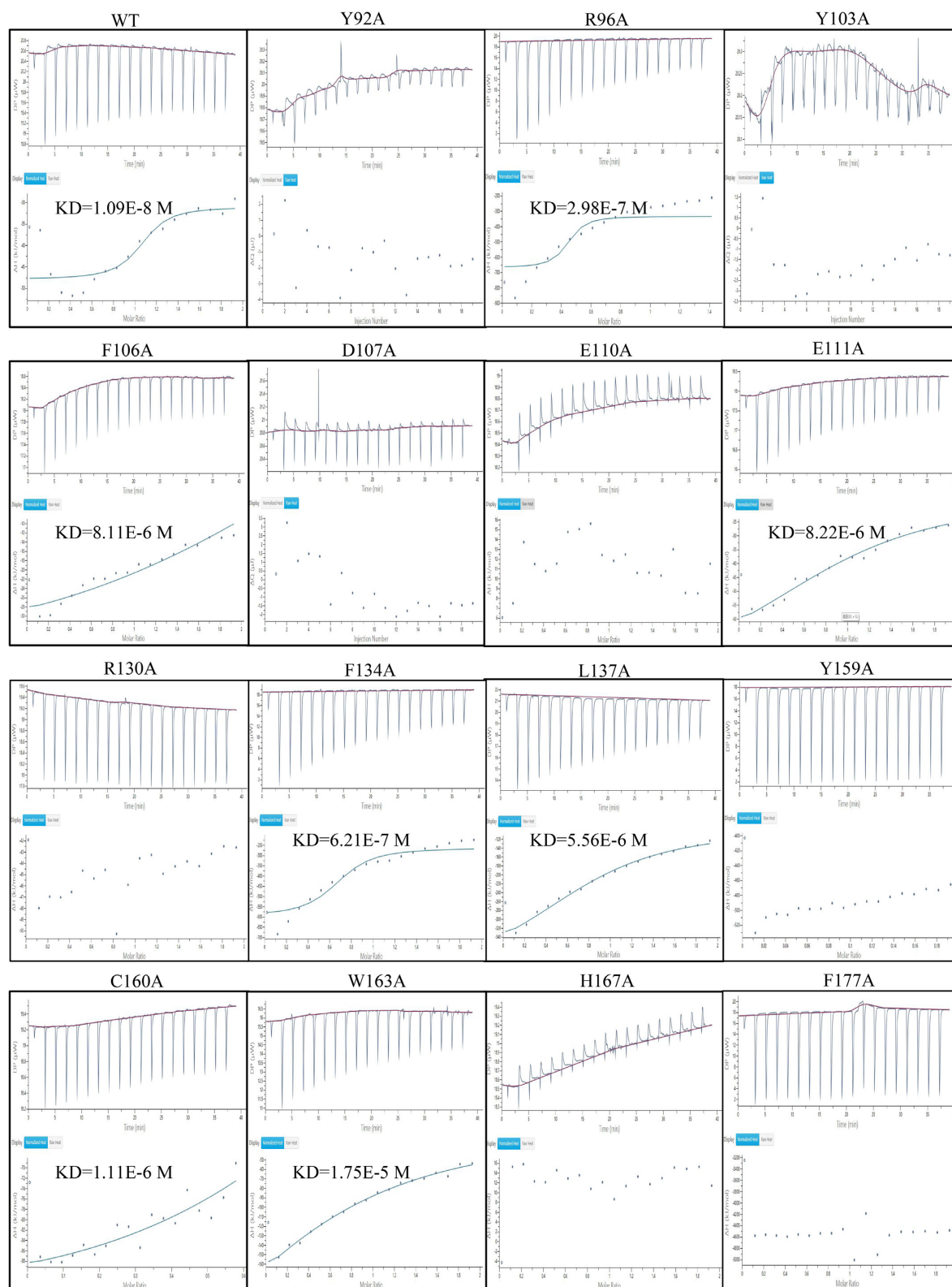


Figure 8. Binding affinities between palmitoyl-CoA and PvrA and the point mutants were measured by ITC. The binding affinities are shown in the panels. No detectable binding was exhibited between palmitoyl-CoA and the Y92A, Y103A, D107A, E110A, R130A, Y159A, H167A or F177A mutant.

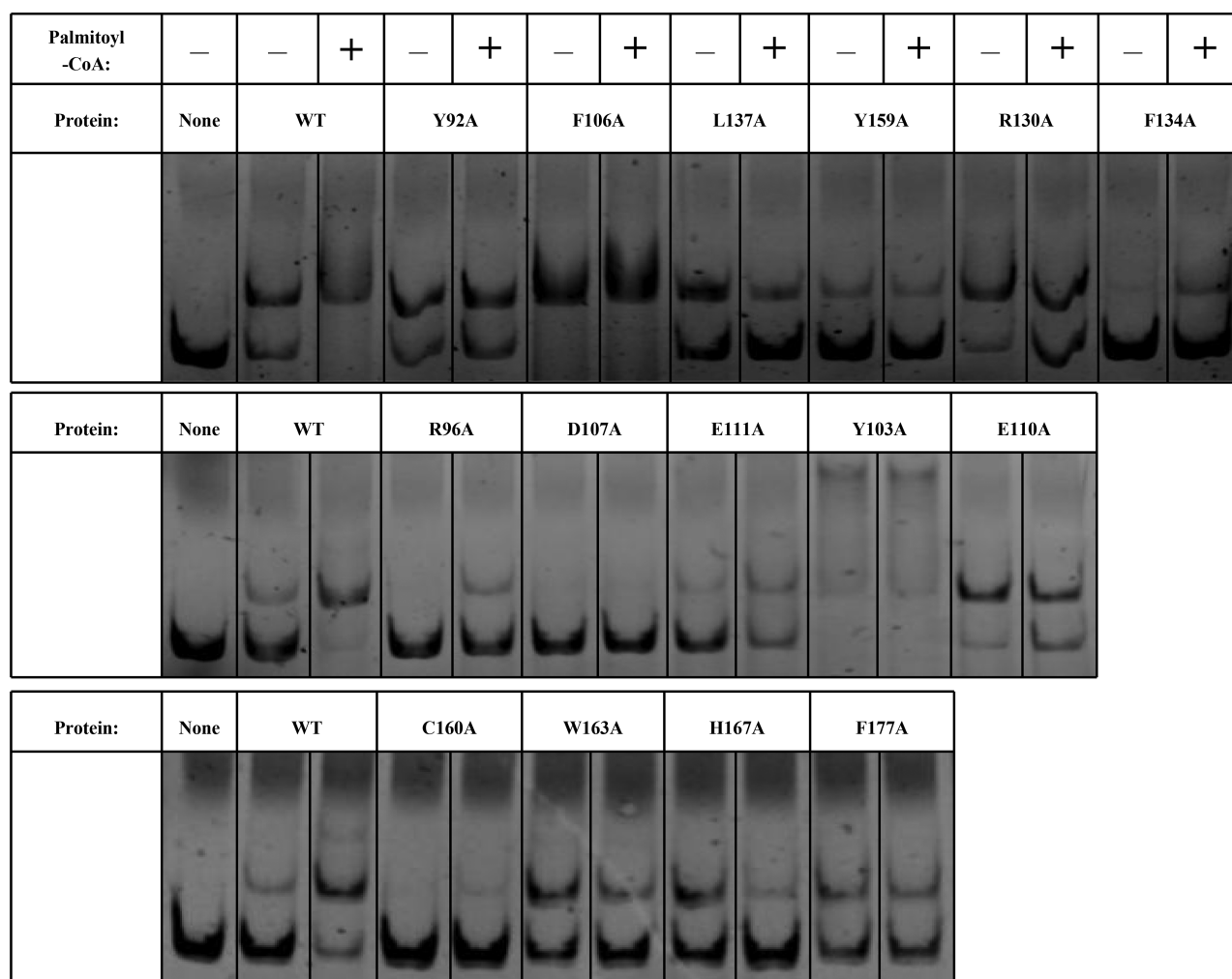


Figure 9. Binding of the wild-type PvrA or indicated point mutants to the *phaG* promoter region in the presence or absence of palmitoyl-CoA was examined by EMSA.

(55). PC is cleaved by *P. aeruginosa*-secreted phospholipase C (PlcH) and lipases, resulting in LCFAs (mainly palmitic and oleic acids), glycerol and phosphorylcholine (34). In *E. coli*, extracellular LCFAs are transported into the cell through a membrane transporter FadL, followed by conversion to fatty acyl-CoA by a fatty acyl-CoA synthetase FadD. The fatty acyl-CoA is then metabolized through β -oxidation and glyoxylate shunt (56–58). Previously, we found that PvrA activates the expression of genes involved in β -oxidation and glyoxylate shunt. Our results in this study expanded the regulatory targets of PvrA to PQS and PHA synthesis genes, revealing the role of PvrA as a global regulator that controls metabolism and virulence genes by binding to host-derived metabolites.

There are two autoinducer molecules in the PQS system, namely HHQ and PQS. HHQ is synthesized by PhnAB and PqsABCDE from chorismate, malonyl-CoA and octanoyl-CoA, which is converted to PQS by PqsH (Figure 11) (59,60). After binding to HHQ or PQS, the transcription regulator PqsR binds to the promoter regions of the *phnAB* and *pqsABCDE* operons and the *pqsH* gene, and activates their expression, thus forming a positive feedback

loop (24,61,62). We found that PvrA activates the production of rhamnolipids through the PQS system in response to palmitic acid and represses the production of PHAs. Rhamnolipids are secondary metabolites that play important roles in bacterial virulence, such as cytotoxicity and involvement in biofilm formation (13,63–65). We further demonstrated a role for rhamnolipids in facilitating the utilization of palmitic acid by *P. aeruginosa*, presumably due to their surface-active property (49). PHAs are polyesters that serve as energy and carbon storage (66,67). In *P. aeruginosa*, the syntheses of rhamnolipids and PHAs share common precursors, i.e. (*R*)-3-hydroxyfatty acids that can be generated from *de novo* fatty acid synthesis or β -oxidation (Figure 11) (68–70). In the synthesis of rhamnolipids, the (*R*)-3-hydroxyacyl-acyl carrier protein (ACP) intermediates are converted by RhlA to (*R*)-3-((*R*)-3-hydroxyalkanoxyloxy) alkanolic acids (HAAs), which are converted to mono-rhamnolipid and di-rhamnolipid by RhlB and RhlC rhamnosyltransferase sequentially (71). In the synthesis of PHA, the (*R*)-3-hydroxydecanoyl moiety of the (*R*)-3-hydroxyacyl-acyl-ACP intermediates are transferred to CoA by 3-hydroxydecanoyl-ACP-CoA transacy-

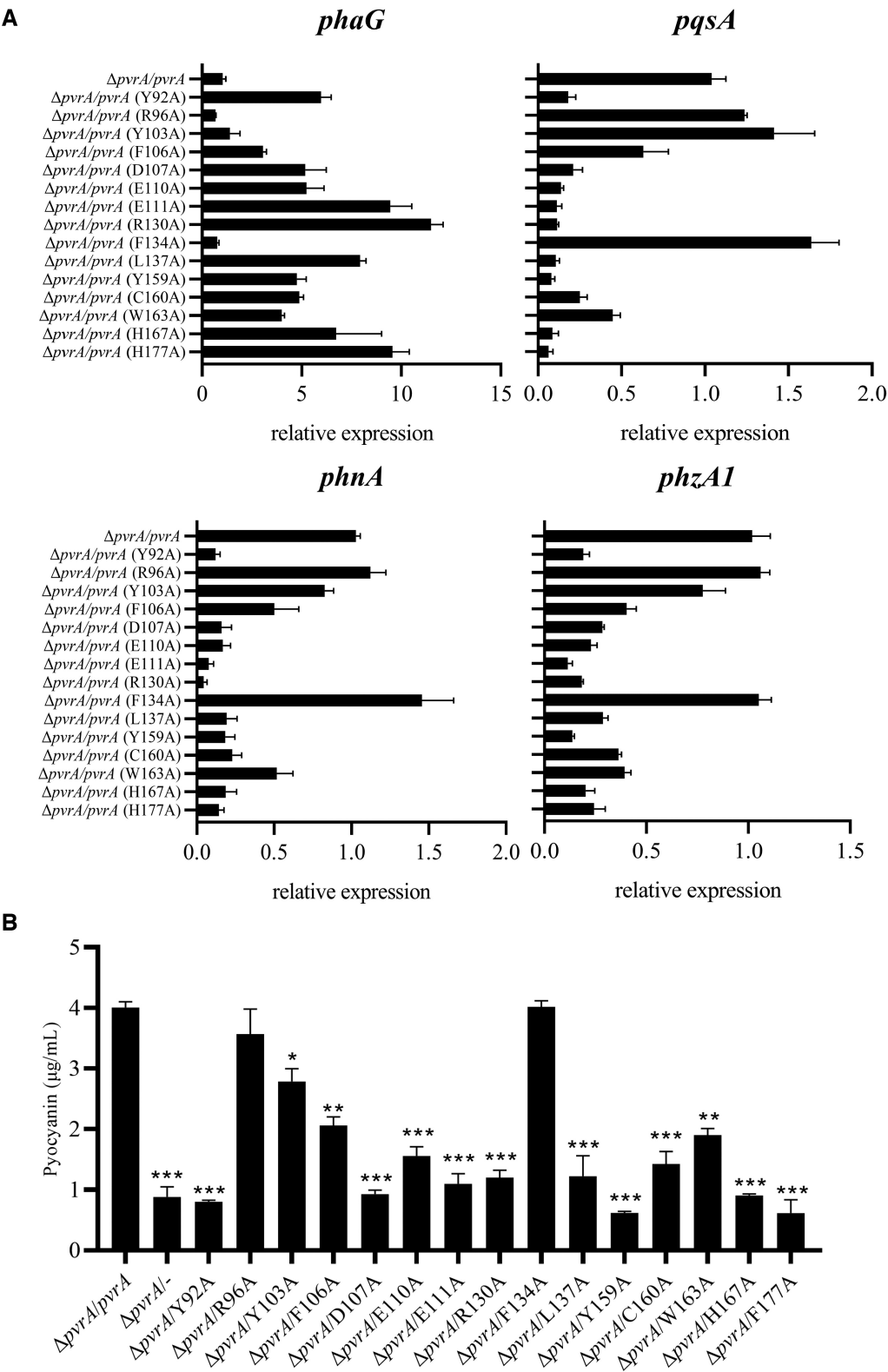


Figure 10. Roles of the potential palmitoyl-CoA-interacting residues in the regulatory functions of PvrA. (A) The $\Delta pvrA$ mutant complemented with wild-type *pvrA* and the indicated point mutants were grown in FA-M9 overnight. The relative mRNA levels of *phaG*, *pqsA*, *phnA* and *phzA* were determined by qRT-PCR. ** $P < 0.01$; *** $P < 0.001$ by Student's *t*-test. Data represent the mean $\pm$ SD from three samples. (B) The $\Delta pvrA$ mutants complemented with wild-type *pvrA* and the indicated point mutants were grown in FA-M9 for 48 h before measurement of pyocyanin yields.

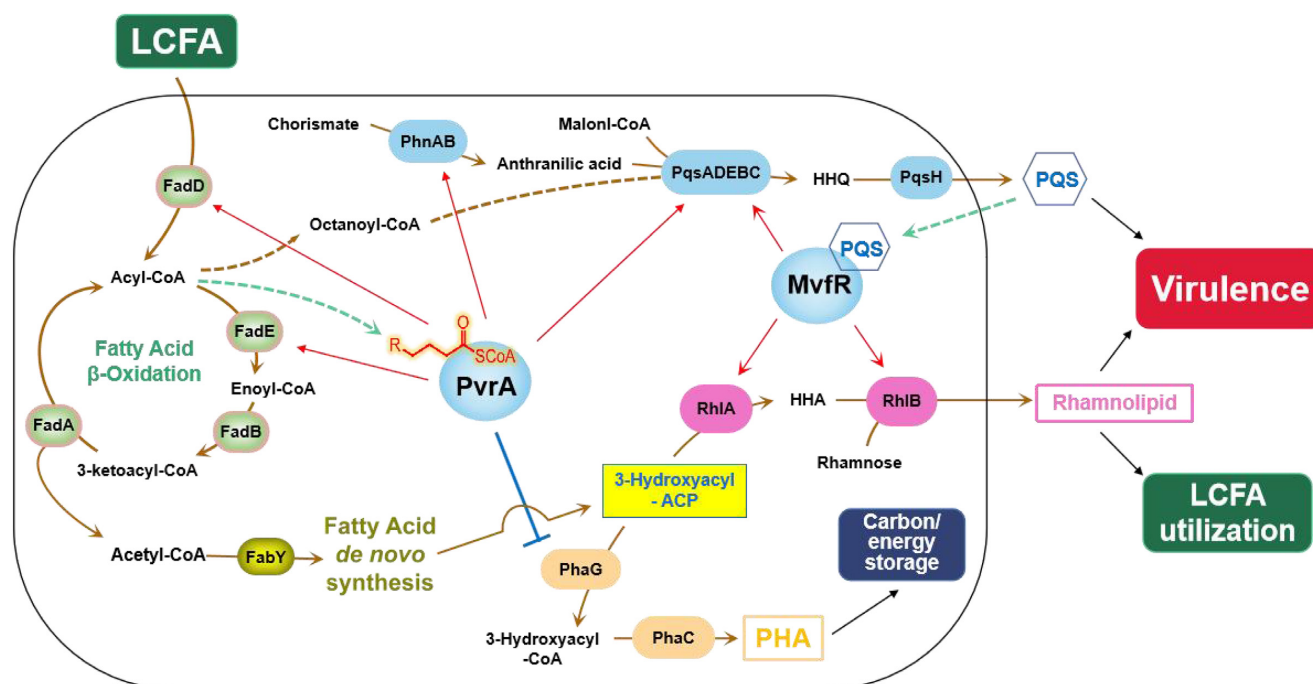


Figure 11. Proposed model for PvrA-mediated regulation on gene expression and carbon flow. LCFAs are converted into fatty acyl-CoA by FadD, and then catabolized through the β -oxidation pathway (27). During each cycle, the fatty acyl-CoA is degraded into a fatty acid chain that is reduced by two carbons and an acetyl-CoA. The octanoyl-CoA generated from the β -oxidation is a substrate for the synthesis of HHQ and PQS (59). Acetyl-CoA serves as the basic building block for the *de novo* biosynthesis of fatty acids, in which 3-hydroxyacyl-ACP is generated. 3-Hydroxyacyl-ACP is catalysed by PhaG and RhlA for the syntheses of PHA and rhamnolipids, respectively (68,70). Meanwhile, PvrA binds to β -oxidation-generated fatty acyl-CoA and activates the expression of genes involved in fatty acid utilization (26) and PQS synthesis (*pqsABCDE* and *phnAB*), while repressing *phaG* expression. Activation of the PQS synthesis genes up-regulates multiple virulence genes, including *rhlA* and *rhlB* for the synthesis of rhamnolipids, which contributes to bacterial pathogenesis and utilization of LCFAs. Repression of *phaG* expression by PvrA blocks the utilization of 3-hydroxyacyl-ACP for PHA synthesis which, together with the up-regulation of *rhlA*, directs carbon flow towards rhamnolipid synthesis.

lase PhaG. Choi *et al.* demonstrated that RhlA can partially substitute the function of PhaG (70). In agreement with the report, we found that mutation of *phaG* slightly reduced the PHA production in FA-M9 (Figure 2A). Collectively, our results demonstrated that by activating the expression of *rhlAB* through PQS and repressing the expression of *phaG*, PvrA directs the energy and carbon flow towards the synthesis of rhamnolipids, thereby facilitating the utilization of host-derived LCFAs and bacterial colonization. The regulatory mechanism and carbon flow pathways are summarized in Figure 11.

PvrA is a TetR family transcriptional regulator that is broadly distributed in prokaryotes (72,73). This family of regulators contain an N-terminal DNA-binding HTH motif and a C-terminal ligand-binding pocket (Figure 7). TetR family proteins act as dimers and bind to DNA sequences with dyad symmetry (72,73). The TetR of *E. coli* represses *tetA* transcription by binding to its promoter region. The binding between tetracycline and the TetR C-terminal pocket releases the repression, leading to up-regulation of TetA and transportation of tetracycline out of the cell (72,73). Like TetR, the majority of the TetR family members function as repressors. However, several TetR family regulators have been reported to function as an activator or as both a repressor and an activator. AtrA of *Streptomyces griseus* activates the transcription of a *strR* gene that encodes the positive regulator of the strepto-

mycin biosynthesis genes (74). CcrR of *Methylobacterium extorquens* activates the transcription of a crotonyl-CoA reductase/carboxylase gene (75). VceR of *Vibrio cholerae* positively regulates its own expression and negatively regulates the *vceCAB* multidrug efflux operon (76). LuxR of *Vibrio harveyi* has been found to directly activate and repress multiple genes (77). PsrA of *Pseudomonas syringae* directly represses the transcription of its own gene and the *aefR* operon that is involved in quorum sensing. Meanwhile, PsrA directly activates the transcription of the alternative sigma factor gene *rpoS* (78).

In our previous and current studies, we found that PvrA directly activates the transcription of *plcH*, *fadD1*, *fadD6*, *fadE*, *glcB*, *maeB*, *pqsA* and *phnA*, and represses the transcription of *phaG* (26). Here we determined the homodimeric structure of PvrA and identified key residues that are involved in the binding of palmitoyl-CoA. In our previous study, R136 was predicted to form hydrogen bonds with the diphosphate group of CoA based on a predicted monomer structure and molecular docking. An R136A mutation reduced the binding affinity between PvrA and palmitoyl-CoA (26). In the PvrA crystal structure, R136 instead forms hydrogen bonds with the side chain of E139. Since L137 is involved in the binding to palmitoyl-CoA (Figure 8), the R136A mutation might perturb the local structure and influence the positioning of L137, thereby reducing the binding affinity. Based on the potential PvrA-

binding sequence (5'-CGGTCA-3') from the ChIP-seq result (26), we identified palindromic sequences in the upstream regions of nine genes, namely *plcH*, *fadD1*, *fadD6*, *fadE*, *glcB*, *maeB*, *pqsA*, *phnA* and *phaG*. Locations and sequences of the PvrA-binding sites are summarized in Supplementary Figure S6D. For the genes that are positively regulated by PvrA, the distances between the two PvrA-binding sites range from 4 to 18 bp, mainly ~6 bp (Supplementary Figure S6D). Further studies are warranted to elucidate the mechanism of PvrA-mediated positive and negative regulation.

In combination, our results demonstrate that PvrA serves as an intracellular sensor for host-derived metabolites and regulates the expression of virulence genes and metabolism pathways.

DATA AVAILABILITY

The RNA-seq data have been deposited in the NCBI with the accession number: PRJNA846955.

Crystallography atomic coordinates and structure factors are deposited in the PDB (<https://www.rcsb.org>) under the following accession codes: 7Y0Y, 7Y0Z and 8HJB.

SUPPLEMENTARY DATA

[Supplementary Data](#) are available at NAR Online.

FUNDING

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Conflict of interest statement. None declared.

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